Grating Pitch Tolerance and TIR Validity for RGB FOV

GUI User Guide

Yen-Chi Chung (Angel)

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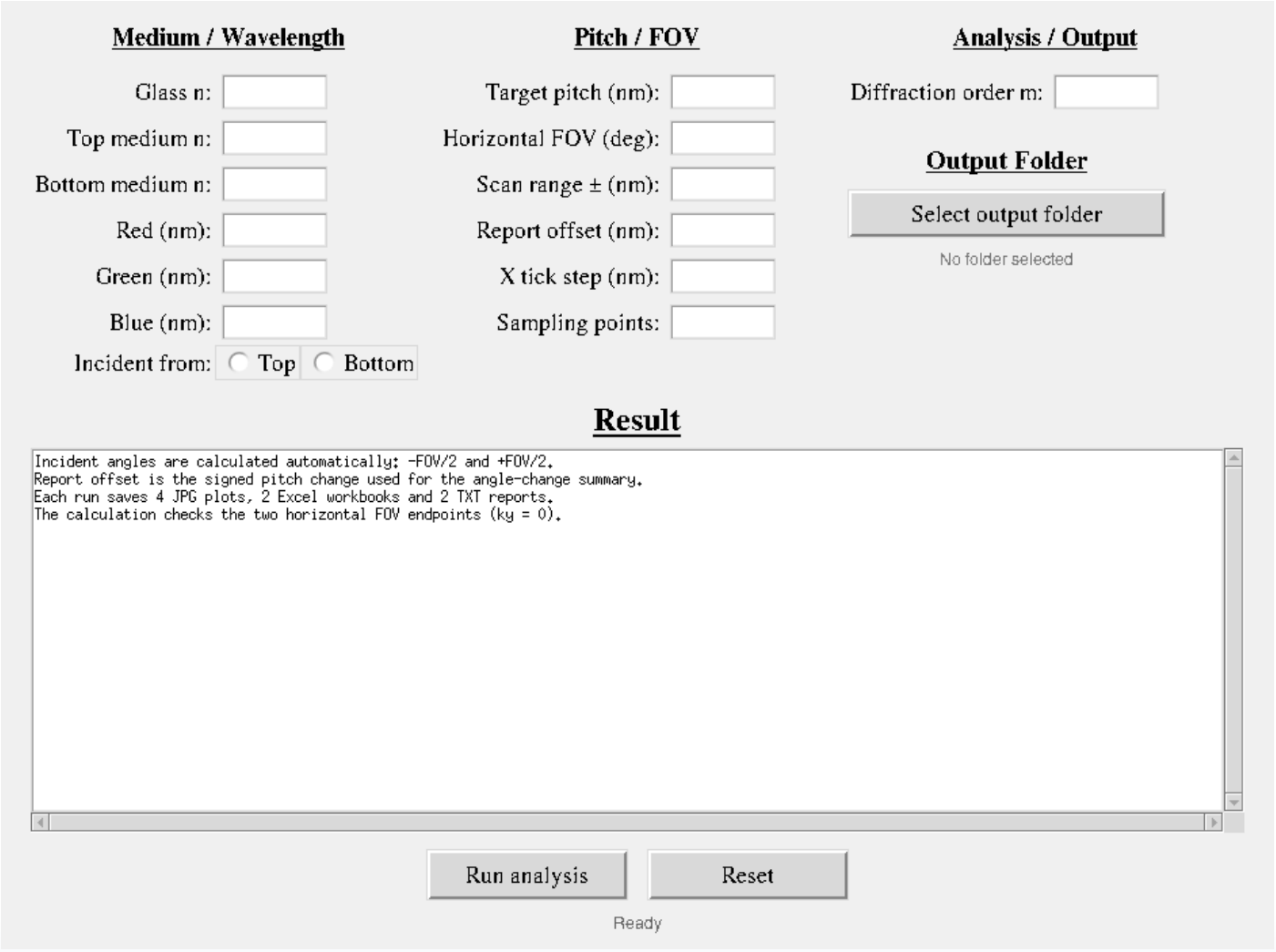
Grating Pitch Tolerance – GUI

Purpose and initial screen

Scan grating pitch and evaluate the resulting diffraction angles.

Analyze −FOV/2 and +FOV/2 separately for RGB.

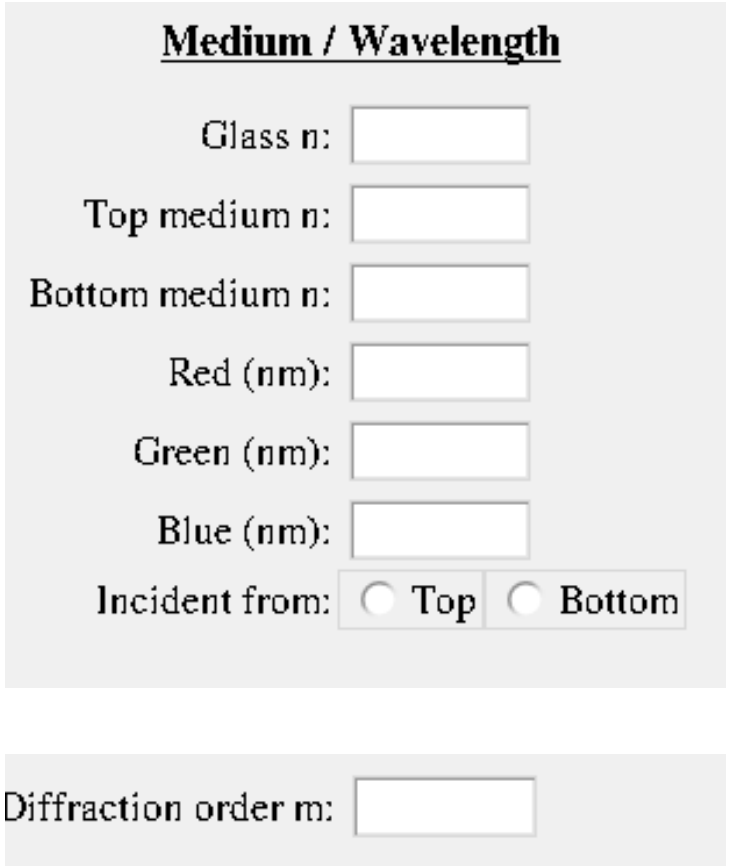
All inputs start blank. Choose Top or Bottom and enter an integer order m.



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Grating Pitch Tolerance – GUI

1. Medium parameters, wavelengths and order



Glass n, Top medium n and Bottom medium n define the optical media.

Top / Bottom selects n_inc automatically.

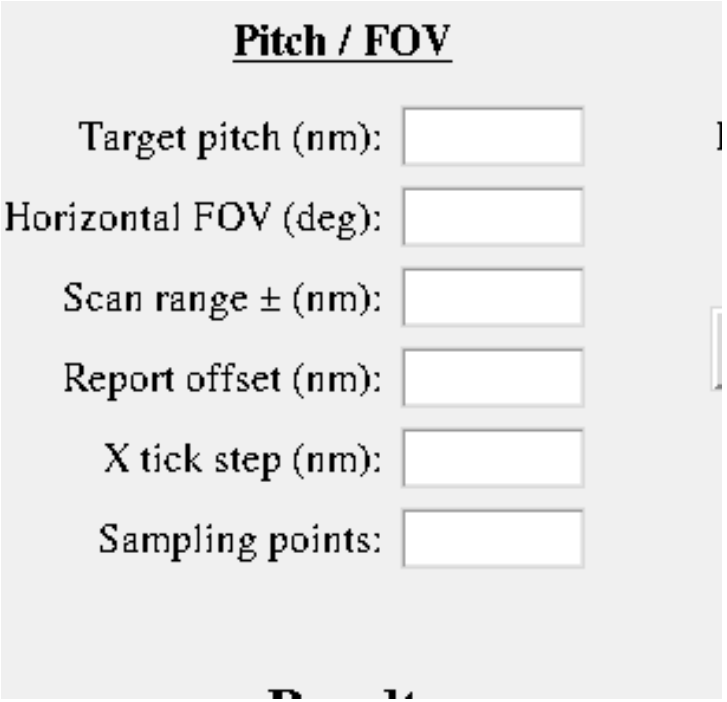
Enter the Red, Green and Blue wavelengths in nm.

Diffraction order m is one signed integer per run, such as −1.

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2. Target pitch, FOV and scan range



Field	Example	Meaning
Target pitch	350 nm	Nominal pitch used as the reference
Horizontal FOV	30°	Analyze input angles −15° and +15°
Scan range ±	50 nm	Scan from 300 to 400 nm

Pitch error = scanned pitch − target pitch

The minimum scanned pitch must stay above zero.

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3. Report offset, tick step and sampling

Field	Purpose	Example
Report offset (nm)	A signed pitch change for the TXT angle-change report	+10 compares 360 nm with 350 nm
X tick step (nm)	Major X-axis tick spacing on the angular-error plot	5 gives a tick every 5 nm
Sampling points	Number of evenly spaced pitch values in the scan	101 points across 300–400 nm gives a 1 nm step

Tick spacing controls the display. Sampling controls calculation density.

The nominal pitch is added if absent, so the actual sample count can exceed the requested count by one.

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Grating Pitch Tolerance – GUI

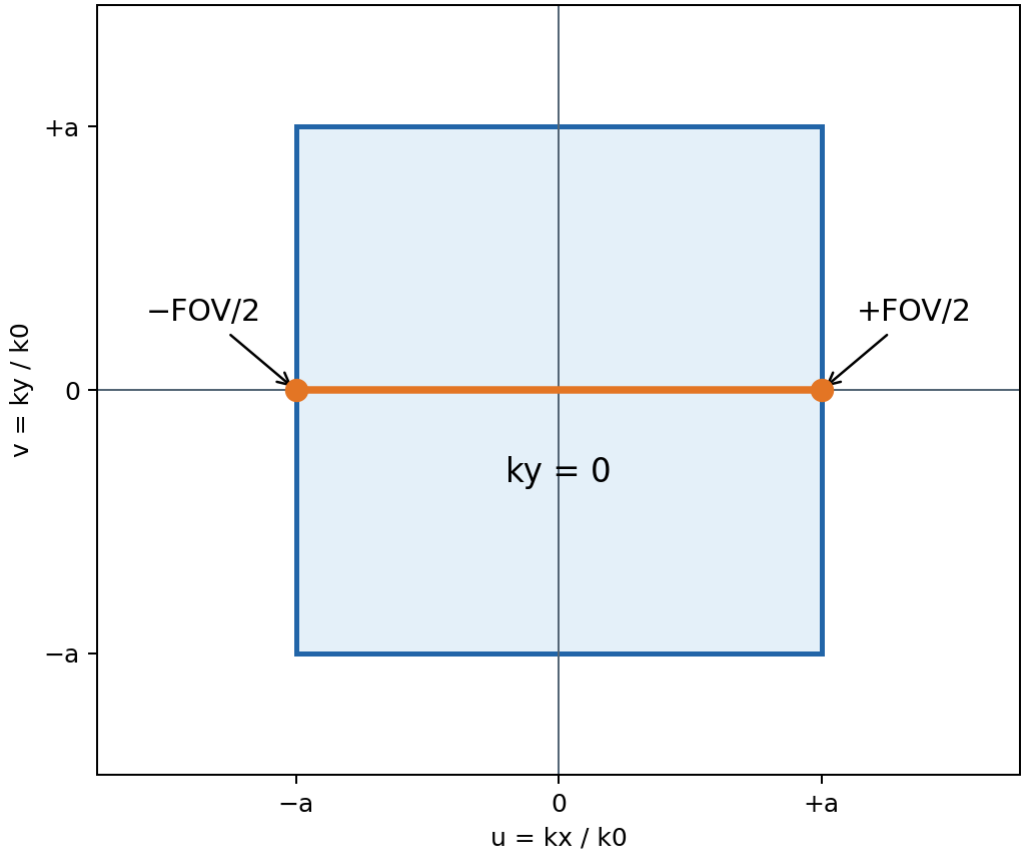
4. Why the input is Horizontal FOV

The calculation fixes ky = 0.

Only the left and right input endpoints are evaluated.

FOV = 30° gives horizontal angles −15° and +15°, with vertical angle 0°.

This does not test the square corners or the entire 2D FOV.



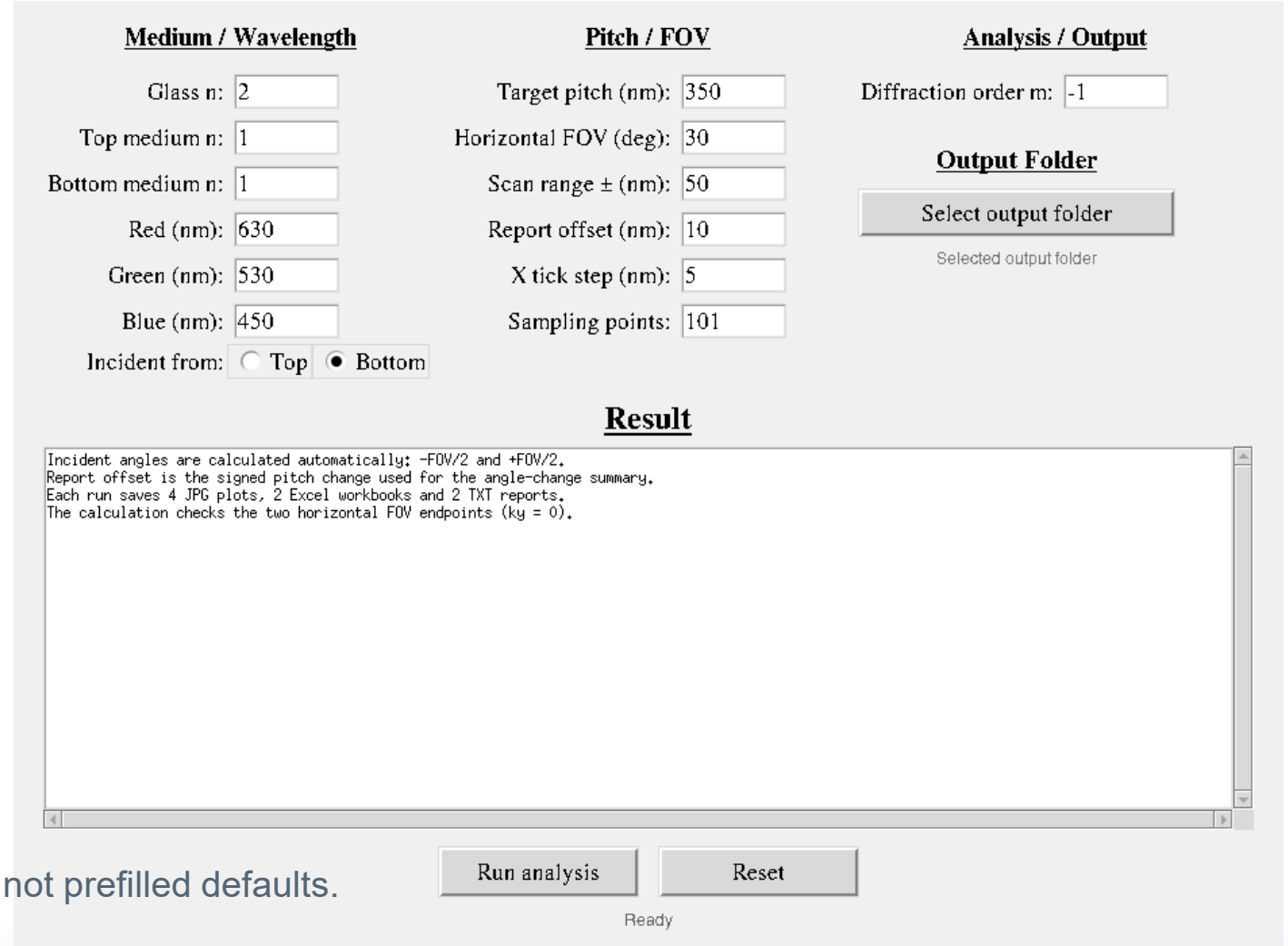
Use the Grating k-space GUI for whole-square FOV validation.

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5. Select output folder and run

1. Fill all numeric fields.
2. Select Top or Bottom.
3. Enter one diffraction order m.
4. Click Select output folder.
5. Click Run analysis.



Example values shown here are entered manually, not prefilled defaults.

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6. Preview the four result plots

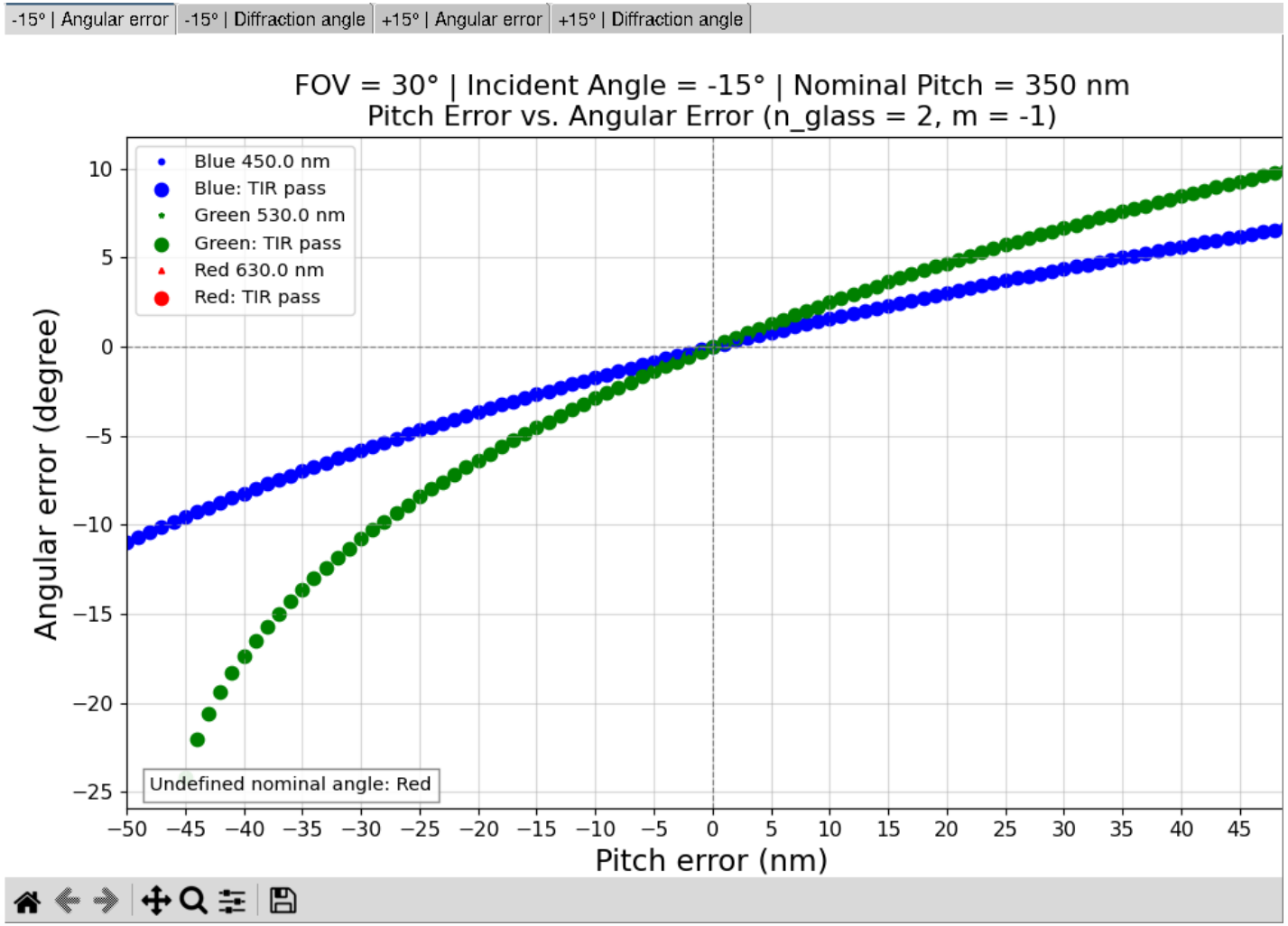
−FOV/2: Angular error

−FOV/2: Diffraction angle

+FOV/2: Angular error

+FOV/2: Diffraction angle

Switch tabs to review each result.
The main GUI shows “Analysis completed.”

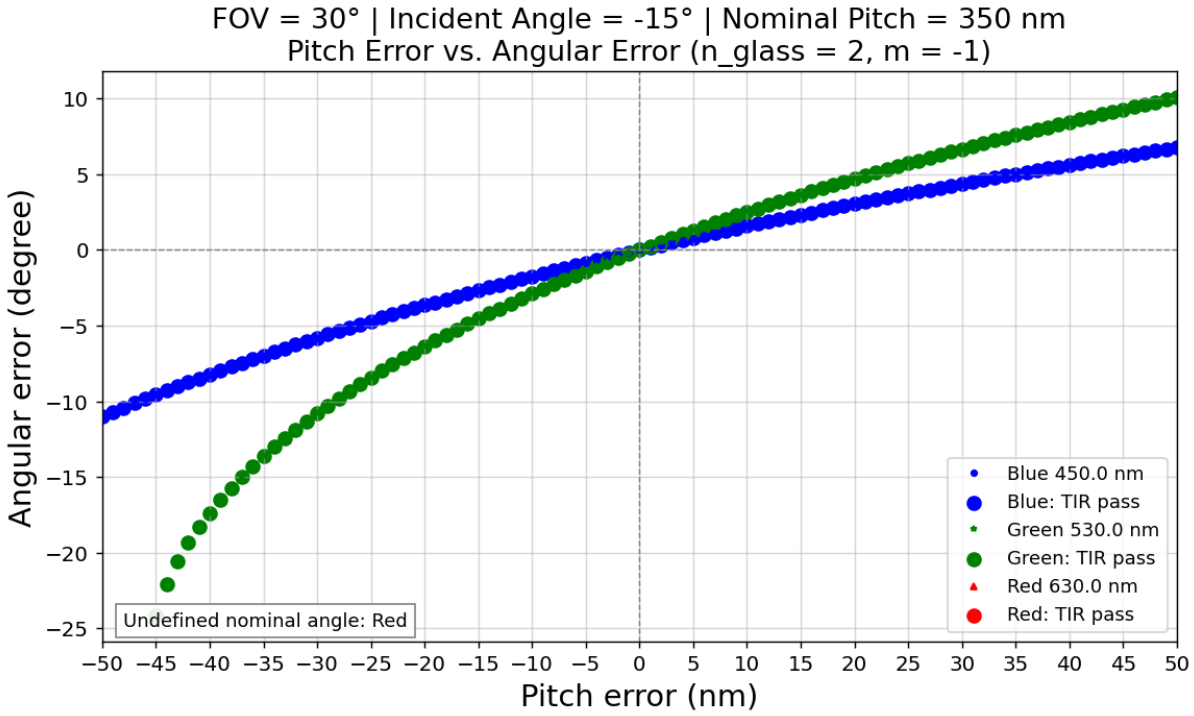


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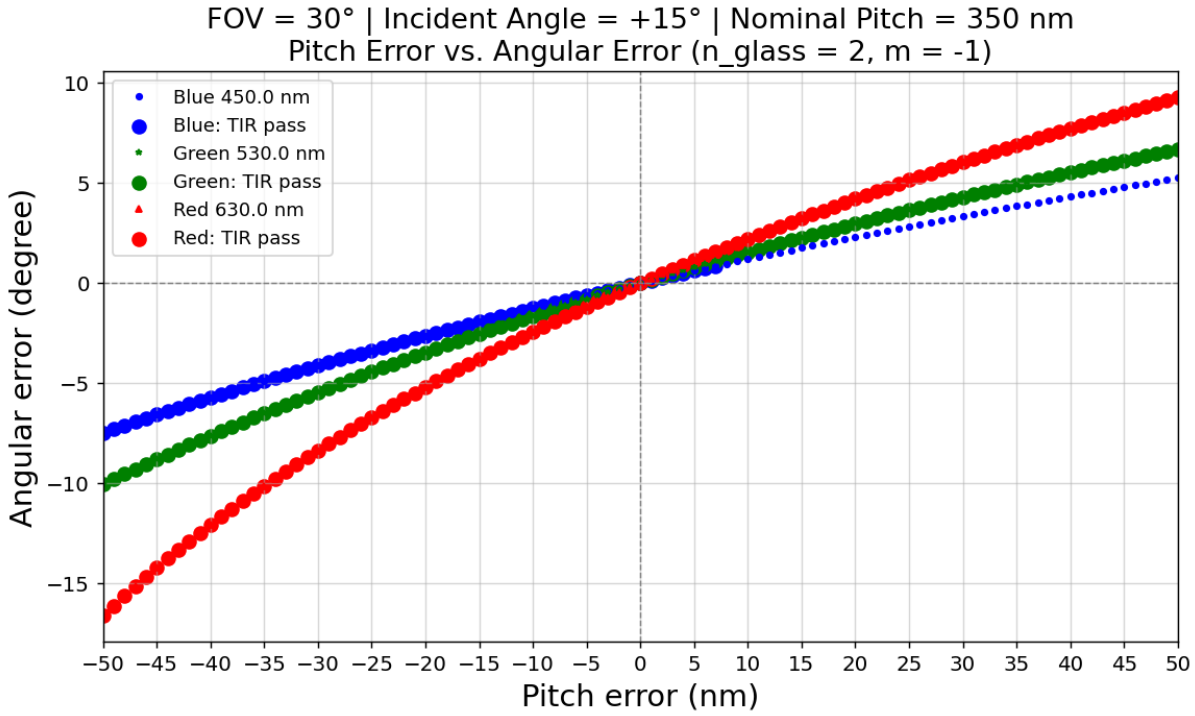
Grating Pitch Tolerance – GUI

7. Pitch error vs. angular error

Input −15°



Input +15°



X: pitch error (nm) Y: θ(pitch) − θ(target pitch)

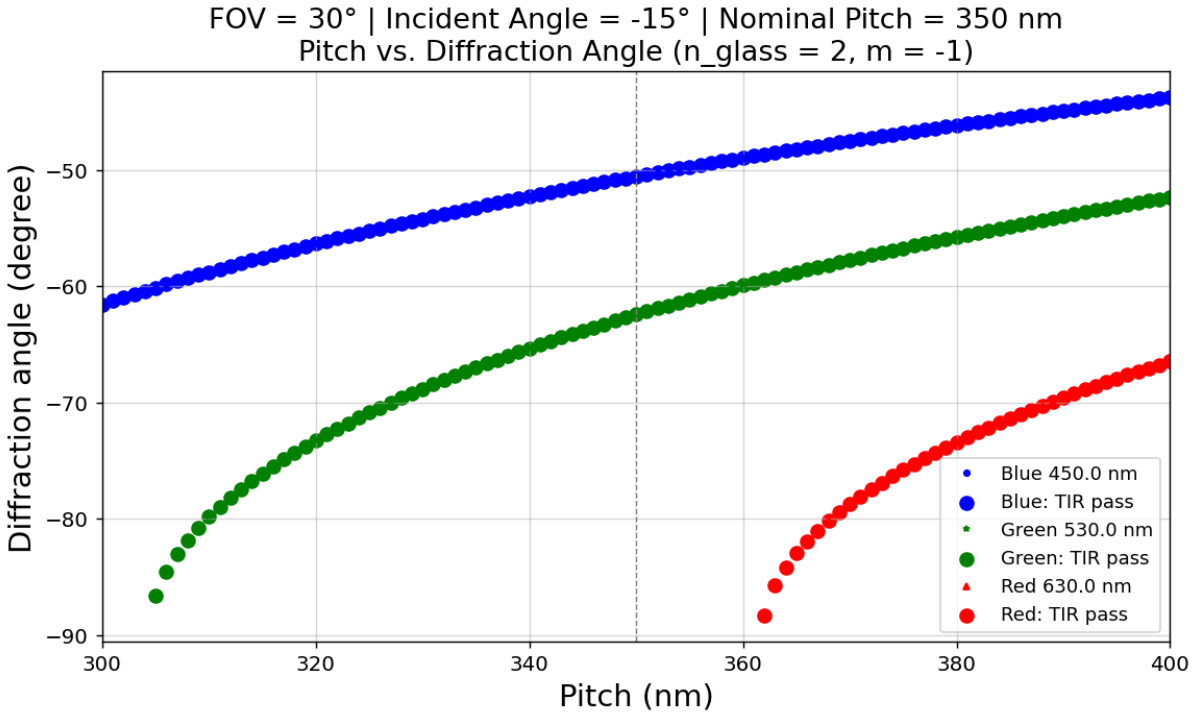
Small markers show real-angle results. Larger dots highlight TIR-pass points with defined angular error.

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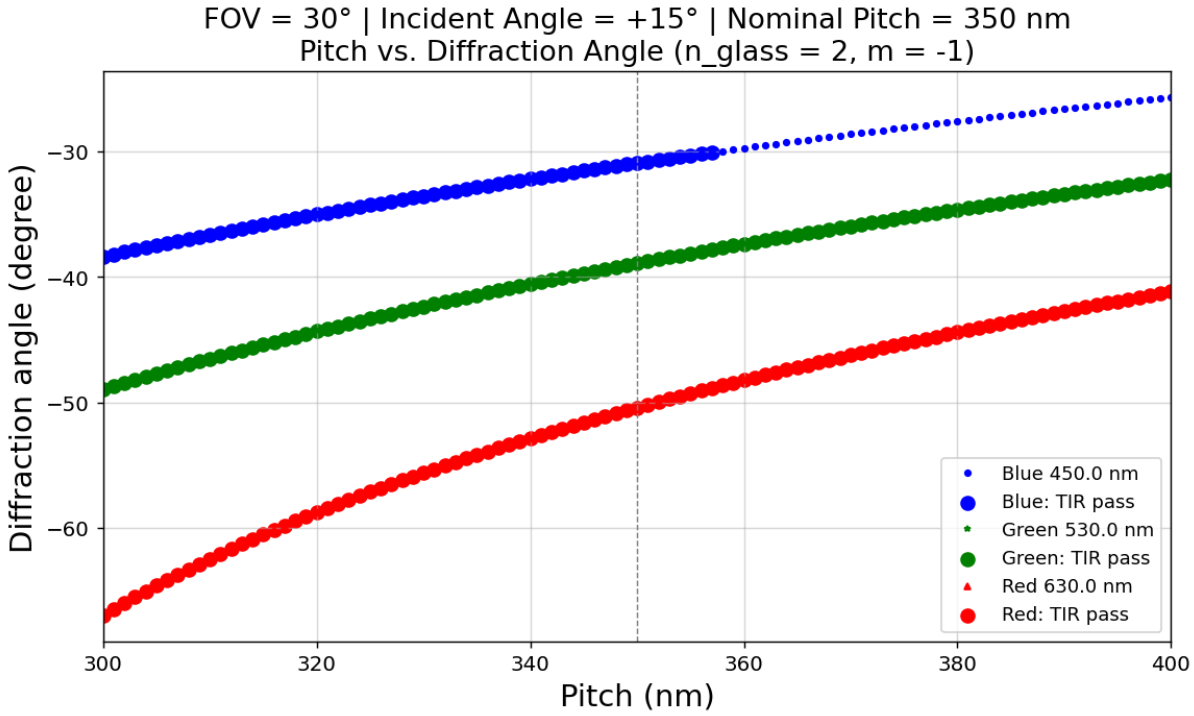
Grating Pitch Tolerance – GUI

8. Pitch vs. diffraction angle

Input −15°



Input +15°



X: actual pitch (nm) Y: signed diffraction angle in glass

Angles are measured from the glass normal. A negative angle indicates the opposite x direction.

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9. TIR validity and missing points

$$u = n_{inc} \sin(\theta_{inc}) + m\lambda / pitch$$

$$\theta_{glass} = \arcsin(u / n_{glass})$$

Condition	Interpretation
max(n_top, n_bottom) < |u| < n_glass	Strict TIR pass at both interfaces
|u| ≤ max(n_top, n_bottom)	Does not satisfy strict TIR at both interfaces
|u| = n_glass	Grazing boundary, not a strict TIR pass
|u| > n_glass	No real propagating glass direction

If the nominal angle is undefined, the angular-error curve is undefined even where other pitches propagate.

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10. Excel output

Column	Meaning
Pitch / Pitch error (nm)	Scanned pitch and its offset from nominal
Diffraction angle (deg)	Signed glass angle at each scanned pitch
Angular error (deg)	Angle relative to the nominal-pitch angle
TIR pass	True / False for the strict TIR test
Status	TIR pass, below threshold, boundary or no propagating direction

Two workbooks per run. Each contains Blue, Green and Red worksheets.

Non-pass entries are red in the angle, TIR-pass and status columns. Undefined angles export as blank cells.

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11. Report offset results in the TXT files

Example: target pitch 350 nm; report offset +10 nm; comparison pitch 360 nm.

Input angle	Blue Δθ	Green Δθ	Red Δθ
−15°	+1.584°	+2.501°	Undefined at nominal pitch
+15°	+1.185°	+1.532°	+2.197°

Open Parameters_<case>.txt and find “Angular Error Analysis”.

The report contains nominal angle, signed angular change and absolute change.

Example: Glass n = 2, surrounding n = 1, RGB = 630 / 530 / 450 nm, m = −1, FOV = 30°.

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12. Output folder and saved files

File prefix	Count	Content
Angular_Error_<case>.jpg	2	Pitch error vs. angular error, one per endpoint
Diffraction_Angle_<case>.jpg	2	Pitch vs. diffraction angle, one per endpoint
Table_<case>.xlsx	2	RGB data worksheets, one per endpoint
Parameters_<case>.txt	2	Settings, model notes and Report offset results

Each run creates a new Pitch_Analysis_<timestamp> subfolder.

Eight files per run. JPG images use 360 dpi. Case names identify input angle, FOV, pitch, Glass n and order.

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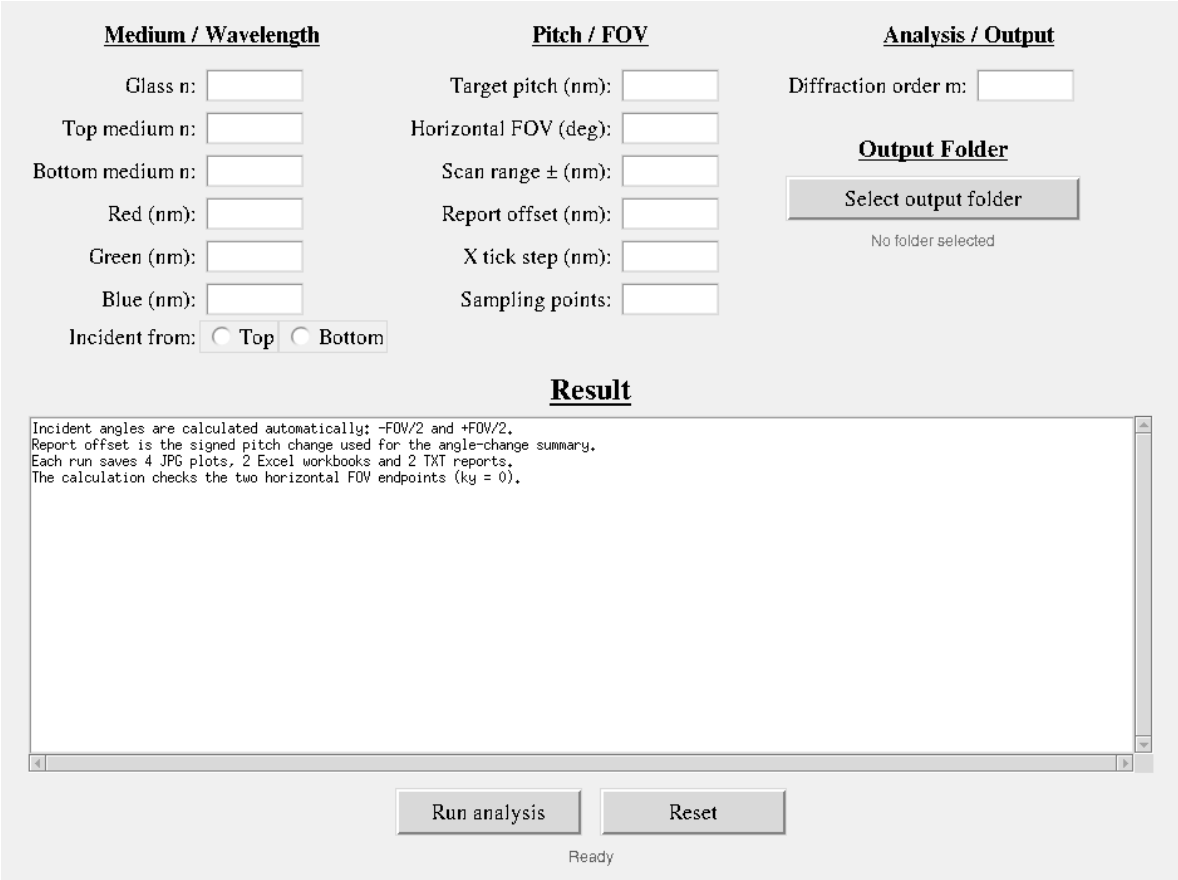
13. Reset and input checks

Reset clears all values, Top / Bottom selection and the output folder.

The preview closes. Previously saved files remain.

Sampling points: integer from 2 to 100,000.
Order m: integer.

Use 0° < horizontal FOV < 180°. Glass n must exceed both surrounding indices.



If the X tick step is too small, increase it to keep the angular-error plot readable.

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Grating Pitch Tolerance – GUI

14. Scope and quick reference

Task	Where to look
Compare pitch sensitivity	Angular-error plots at both horizontal endpoints
Check directional TIR validity	Large plot markers and Excel TIR pass / Status
Read a specific pitch-offset result	TXT: Angular Error Analysis
Check the full 2D FOV	Use the Grating k-space GUI
Evaluate coupling efficiency	Requires a separate electromagnetic calculation

Launch: python Pitch_Deviation_GUI.py | Dependencies: numpy, matplotlib, pandas, openpyxl and Tkinter

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